

Problem Set 1

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Problem 1

A random variable X takes values $\{1, 2, 4\}$ with probabilities $\{0.2, 0.5, 0.3\}$:

$$P(X = 1) = 0.2, \quad P(X = 2) = 0.5, \quad P(X = 4) = 0.3$$

a)

Compute the expected value $\mathbb{E}[X]$.

The formula for expected value is $\mathbb{E}[X] = \sum x_i P(X = x_i)$

The expected value for X is

$$\mathbb{E}[X] = \sum_{x_i} x_i P(X = x_i) = 1 * 0.2 + 2 * 0.5 + 4 * 0.3 = 2.4$$

b)

Explain what the expected value represents in this context.

The expected value is the weighted sum of all possible outcomes, where the weight of each outcome is its respective probability.

Problem 2

Using the random variable X from Problem 1:

a)

Compute $\mathbb{E}[X^2]$.

$$\mathbb{E}[X^2] = \sum x_i^2 P(X = x_i) = 1 * 0.2 + 4 * 0.5 + 16 * 0.3 = 8.8$$

b)

Compute the variance $Var(X)$.

The formula for variance is $Var(X) = \mathbb{E}[(X - \bar{X})^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

$$\mathbb{E}[X^2] - \mathbb{E}[X]^2 = 8.8 - 2.4^2 = 3.04$$

c)

Compute the standard deviation σ_X .

The standard deviation is the square root of the variance:

$$\sqrt{Var(X)} = \sqrt{(3.04)} \approx 1.744$$

d)

Interpret the meaning of the standard deviation.

The standard deviation measures dispersion about a mean. Higher values indicate greater dispersion.

Problem 3

a)

A trader receives \$1 if a stock goes up tomorrow and \$0 otherwise. The probability the stock goes up is $p = 0.6$.

b)

Define the random variable X .

$$P(X) = \begin{cases} 0.4, & X = \$0 \\ 0.6, & X = \$1 \end{cases}$$

c)

Identify the distribution of X .

The distribution of X is discrete, binary. This is a Binomial distribution over one trial. A Binomial distribution with one trial is a Bernoulli distribution.

d)

Compute $\mathbb{E}[X]$, $\text{Var}(X)$, and σ_X .

The expectation, variance, and standard deviation of a Bernoulli distribution are as follows:

$$\mathbb{E}[X] = p$$

$$\text{Var}(X) = p(1 - p)$$

$$\sigma_X = \sqrt{p(1 - p)}$$

With $p = 0.6$, we have:

$$\mathbb{E}[X] = 0.6$$

$$\text{Var}(X) = 0.6 * 0.4 = 0.24$$

$$\sigma_X = \sqrt{0.24} \approx 0.490$$

Problem 4

A salesperson makes $n=5$ independent sales calls, each with probability of success $p = 0.3$.

a)

Define the random variable X .

$$X \sim B(n = 5, p = 0.3)$$

X is the number of successful sales calls in a trial of 5 independent sales calls. The distribution of X successes in n independent trials, with probability of success, p , is a Binomial Distribution.

b)

Identify the distribution of X .

The general formula for k successes in n independent trials, with probability of success, p is:

$$Pr(X = k) = \binom{n}{k} p^k * (1 - p)^{n-k}$$

With $p = 0.3$, this evolves into:

$$\begin{aligned} Pr(X = k) &= \binom{n}{k} 0.3^k * 0.7^n * 0.7^{-k} \\ &= \binom{n}{k} 0.7^n * e^{-k(\ln(0.7) - \ln(0.3))} \\ &= \binom{n}{k} 0.7^n * e^{-0.847k} \end{aligned}$$

With $n = 5$, this evolves into:

$$\begin{aligned} \binom{5}{k} 0.7^5 * e^{-0.847k} &= \binom{5}{k} 0.168e^{-0.847k} \\ &= \frac{5! * 0.168e^{-0.847k}}{k!(5-k)!} \\ &= \frac{120 * 0.168e^{-0.847k}}{k!(5-k)!} \\ &= \frac{20.16e^{-0.847k}}{k!(5-k)!} \end{aligned}$$

Tabulated over the 6 possible outcomes, with 3 significant figures:

k	P(X=k)
0	0.168
1	0.360
2	0.308
3	0.132
4	0.028
5	0.002

c)

Compute $P(X = 3)$.

$$P(X = 3) \approx 0.132$$

d)

Compute $\mathbb{E}[X]$ and $\text{Var}(X)$.

The expectation for a binomial distribution is $\mathbb{E}[x] = np$. With $n = 5$ and $p = 0.3$, we have $\mathbb{E}[x] = 1.5$.

The variance for a binomial distribution is $\text{Var}(X) = np(1 - p)$. With $n = 5$ and $p = 0.3$, we have $\text{Var}(X) = 1.05$.

Problem 5

Using the binomial model from Problem 4:

a)

What does the standard deviation tell you about the number of successful calls?

The standard deviation measures the expected dispersion around the mean.

In this case, the s.d. is 1.025, so you may expect 1 to 2 out of 5 calls to succeed, most of the time.

b)

Would you expect most outcomes to be close to the expected value? Why?

The expected value is the projected average over many calls. The standard deviation of 1.025 around the expected value of 1.5 indicates that most trials will see 2-3 successes. This seems likely, given that 2 and 3 successes are the most probable of outcomes in a trial of 5 calls.

Problem 6

The number of customer complaints arriving per hour follows a Poisson distribution with mean $\lambda = 2$ complaints/hour.

a)

Write the probability mass function.

The probability mass function for a Poisson distribution is:

$$Pr(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

With $\lambda = 2$, this evolves into:

$$Pr(X = k) = \frac{2^k e^{-2}}{k!}$$

b)

Compute $P(X = 3)$.

Substituting $k = 3$, we have

$$Pr(X = 3) = \frac{2^3 e^{-2}}{3!} \approx 0.066$$

c)

Compute $\mathbb{E}[X]$, $Var(X)$, and σ_X .

For a Poisson distribution, we have the following formulas for expectation, variance, and standard deviation:

$$\mathbb{E}[X] = \lambda$$

$$\text{Var}(x) = \lambda$$

$$\sigma_X = \sqrt{\lambda}$$

For $\lambda = 2$, this gives us:

$$\mathbb{E}[X] = 2$$

$$\text{Var}(x) = 2$$

$$\sigma_X = \sqrt{2}$$

d)

Explain why the Poisson distribution is appropriate here.

The Poisson distribution is appropriate for i.i.d. events that happen with relative infrequency over a continuum, such as a time, or space. Complaints can be assumed to be independent, and their occurrence at a call center is measured over time.

Problem 7

Compare the variance and expected value of a Poisson distribution.

0.1 a)

What special property do you observe?

The expected value and the variance of a Poisson distribution are the same.

0.2 b)

How does this differ from the binomial distribution?

With the binomial distribution, the expectation and the variance differ by a factor of $(1 - p)$. Furthermore, the lambda term is not applicable for the binomial distribution, which is taken over a discrete event space.

Problem 8

Answer briefly:

0.3 a)

When should a Bernoulli distribution be used instead of a binomial?

A Bernoulli distribution can be used when a sample consists of a single trial.

0.4 b)

Why does the Poisson distribution not require a fixed number of trials?

The Poisson distribution assumes that independent events are happening in a continuous space.

Implicitly, the Poisson distribution is measured over a single unit of the continuum, ex. complaints received in one hour.

The Poisson distribution can be expressed more generally, for a given interval, as such:

$$P(k \text{ events in interval } t) = \frac{(rt)^k e^{-rt}}{k!}$$

We can infer that $\lambda = rt$, and in the simple case, $t = 1$.

The Poisson distribution does not require a fixed number of trials, but it is measured over a fixed interval.

0.5 c)

In practical terms, what does a high standard deviation imply?

A high standard deviation implies a lot of dispersion away from the mean. It may indicate that a particular model is not a good fit for the data.